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AI-BASED INDUSTRIAL MAINTENANCE AND PROCESS OPTIMIZATION

An AI-powered system for predicting machine failures
and optimizing industrial maintenance decisions

Problem Statement

Problems:

Traditional maintenance approaches often depend on:

- Fixed maintenance schedules
- Manual inspection
- Reactive maintenance after machine failure
- Large amounts of sensor data that are difficult to interpret manually

Consequences

- Unexpected machine downtime
- Increased maintenance cost
- Production losses
- Equipment damage
- Difficulty identifying high-risk machines early

Our Solution

We developed an AI-Based Industrial Maintenance system that analyzes machine sensor parameters, predicts failure probability, classifies machine risk, and provides actionable maintenance recommendations.

Project Objectives

1. Predict machine failure before it occurs
2. Calculate **failure probability**
3. Categorize machines into:
 - LOW
 - MODERATE
 - HIGH
 - CRITICAL
4. Compare multiple ML and Deep Learning models
5. Identify important machine risk factors
6. Provide maintenance recommendations
7. Store and analyze prediction history
8. Provide an interactive industrial dashboard
9. Secure user access using authentication

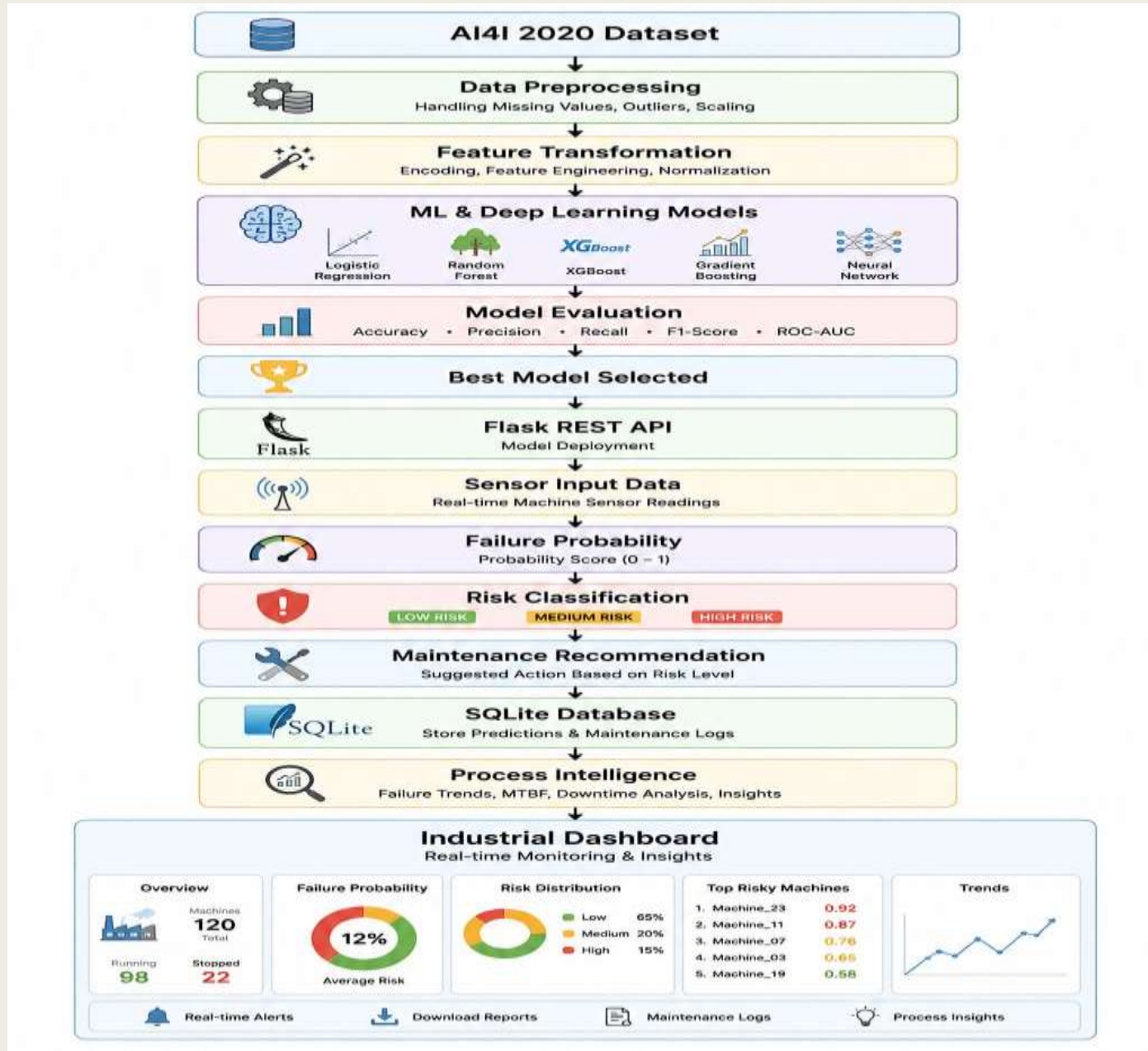
Dataset

AI4I 2020 Predictive Maintenance Dataset

Input Parameters

Feature	Description
Type	Machine variant — L, M, H
Air Temperature	Ambient temperature
Process Temperature	Operating process temperature
Rotational Speed	RPM
Torque	Mechanical load
Tool Wear	Tool usage time

Proposed System



Machine Learning Methodology

Traditional Machine Learning

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost

Deep Learning

Multi-Layer Perceptron (MLP) using PyTorch

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

Model Performance

Metric	Score
Accuracy	98.5%
Precision	83.3%
Recall	64.3%
F1-Score	71.0%
ROC-AUC	93.4%

Key Result:
98.5% Accuracy with **93.4% ROC-AUC**, providing effective machine failure prediction.

Prediction & Risk Assessment

Machine Sensor Inputs

- Machine Type
- Air Temperature
- Process Temperature
- Rotational Speed (RPM)
- Torque
- Tool Wear

AI Prediction :Failure Probability (%)

Risk Classification

Failure Probability	Risk Level
0–30%	LOW
30–60%	MODERATE
60–80%	HIGH
80–100%	CRITICAL

Process Optimization & Analytics

Process Optimization

- Failure Probability Trends
- Machine Risk Distribution
- Machine Variant Distribution
- Average Risk by Machine Type
- High-Risk Machine Alerts
- Historical Prediction Analysis

Key Benefit

- **Transforms individual AI predictions into actionable operational insights for maintenance planning.**

System Features & Dashboard

Dashboard Overview

- Total Evaluations
- Healthy Machines
- At-Risk Machines
- Average Failure Risk
- Machine Health Gauge

Prediction

Sensor Input → AI Prediction → Risk Level → Recommendation

Analytics

- Process Intelligence Charts
- High-Risk Alerts
- Machine Risk Analysis

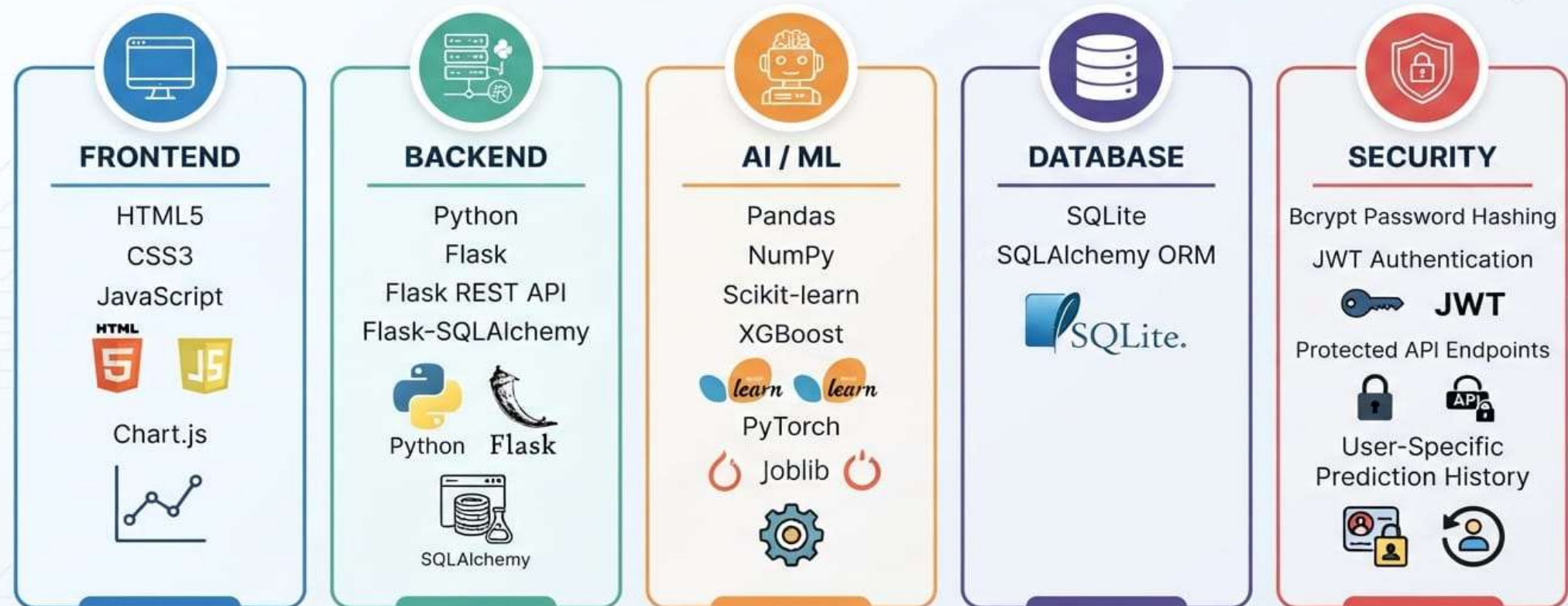
History

- * Searchable Prediction Records
- * Prediction Audit History

Technology Stack & Security

TECHNOLOGY STACK & SECURITY

A Comprehensive Architecture for AI-Driven Predictive Maintenance



System Architecture Overview

Conclusion & Future Scope

Conclusion

An end-to-end AI-Based Industrial maintenance system for early failure detection, risk assessment, and intelligent maintenance decisions.

Future Scope

- Real-Time IoT Sensor Integration
- Cloud Deployment
- Real-Time Machine Monitoring
- Automated Maintenance Scheduling
- Remaining Useful Life (RUL) Prediction
- ERP/CMMS Integration
- Continuous Model Retraining